

USBSID-Pico PCB firmware manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give **no guarantees** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

Firmware types



Each PCB revision is the firmware designator, please check your PCB revision, this is under the MOS logo and next to **USBSID-Pico** on your PCB.

Firmware filenames contain the PCB revision number e.g. **usbsidpico-rgb-v1.X.uf2** is for PCB revision **v1.0** and for use with a black clone Pico containing an RGB LED.

Firmware naming

Below are the filenames used for each Pico and board type, take great care in selecting the correct file for you Pico!

WARNING! Do **NOT** use the **RGB** firmware for any of the (non black) rp2040 or (non black) rp2350 Pico boards that do not contain an RGB LED, this can break your Pico!

Board revision

The 1.X in each filename equals for the PCB revision as mentioned under firmware types. Don't worry if you flash a firmware for the incorrect PCB revision, this causes no harm.

Firmware naming

usbsidA-BC.uf2 examples: **usbsidpico-v1.0-rgb.uf2**, **usbsidpico2_w-v1.3.uf2** or **usbsidpico2-v1.5-rgb_em**.

A: The pico board type e.g. **pico**, **pico2**, **pico_2** or **pico2_w**

B: The PCB version e.g. `v1.0`, `v1.3` or `v1.5`

C: A combination of optional features e.g. `-rgb`, `_em` or `_net`

Firmware naming per Pico type

Firmwares suitable for rp2040 Pico1 boards start with `usbsidpico-` Firmwares suitable for rp2040 Pico1_w boards start with `usbsidpico_w-`

Firmwares suitable for rp2350 Pico2 boards start with `usbsidpico2-`

Firmwares suitable for rp2350 Pico2_w boards start with `usbsidpico2_w-`

Naming examples for the rp2040 Pico1

Firmware naming suited for the pico 1 always start with `usbsidpico-`

`usbsidpico-v1.X.uf2` for regular green rp2040 Pico boards.

`usbsidpico-rgb-v1.X.uf2` for black clone rp2040 Pico boards with RGB LED onboard.

`usbsidpico_w-v1.X.uf2` for regular green rp2040 PicoW boards.

NOTE! Due to the hardware limitations of the RP2040, highly demanding, cycle-exact SID tunes (dense per-cycle register writes, e.g. some digi/sample tunes) may not play at full speed on Pico1 or Pico1_w boards. Use a Pico2 or Pico2_w (rp2350) board for these tunes.

Naming examples for the rp2350 Pico2

Firmware naming suited for the pico 2 always start with `usbsidpico2-`

`usbsidpico2-v1.X.uf2` for regular green rp2350 Pico2 boards.

`usbsidpico2-rgb-v1.X.uf2` for black clone rp2350 Pico2 boards with RGB LED onboard.

`usbsidpico2_w-v1.X.uf2` for regular green rp2350 Pico2W boards.

Firmwares with embedded USBID-Player and Cynthcart

Firmwares ending with `_em` have an embedded USBID-Player that can play PSID and RSID tunes and PSID64 PRG/P00 files, and can also start Cynthcart via CC commands. Both now run through the same embedded player core.

For example `usbsidpico-v1.0_em.uf2`, `usbsidpico2-v1.3-rgb_em.uf2` or `usbsidpico2_w-v1.5_em_net.uf2`

This USBID-Player is still in active development and some tunes might not work or play at all. The embedded player needs a large chunk of RAM (over 80% on rp2040 boards, around 40% on rp2350 boards) and an overclocked CPU, but is now built for every Pico type, rp2040 included. On pico2_w it is only ever offered bundled with Network SID Device support (`_em_net`), since that board has RAM to spare for both together.

WARNING! Whenever the embedded player or Cynthcart is used the Pico MCU is overclocked to 250MHz, regardless of Pico type. Passive cooling `_might/could` be needed.

See the `USBID-Player-manual` and `USBID-Pico-Cynthcart-manual` files for more information.

How to flash your Pico

NOTE:

When flashing a new firmware version, all previously configured settings will be reset to default. Use the commandline configtool to save your current settings to a `ini` file if you want to save them!

A Raspberry Pi Pico board is incredibly easy to flash, as it comes with a built in bootloader for flashing new firmwares in the `uf2` format.

In order to flash a new firmware to your USBID-Pico you will need to put the Pico into bootloader mode.

This can be done in multiple ways:

With the cable already plugged into your computer and Pico and seated on the board (using buttons):

- Press and hold the `BOOTSEL` button on the Pico.
- Press and release the `RST` button on the USBID-Pico board.
- Now release the `BOOTSEL` button.
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

With the cable not plugged in to your computer and Pico and also not seated on the board (using buttons):

- Plug in the USB cable to your Pico and not into your computer.
- While holding the `BOOTSEL` button on the Pico plugin the other end of the USB cable into your computer.
- Now release the `BOOTSEL` button.
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

With the cable already plugged into your computer and Pico and seated on the board (using software):

- With the GUI config tool
- Go to the `DEBUG` tab and click on `BOOTLOADER`
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

or

- Go to the config webpage (usbsid.loudai.nl/index.html?player=webusb&debug=usbsidpico):
- Connect to the board using **WebUSB (Hermit)** as emulator
- Click on **CONFIG** to open the configuration tab
- Scroll to the bottom of the page and click on **BOOTLOADER** in the box named **DEBUG**
- A new drive should appear on your computer called **RPI-RP2** or **RP2350** depending on your Pico type.
- Copy the correct **uf2** firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

or

- With the CLI config tool command in your **PATH** variable
- Type **cfg_usbsid -boot** to send the reboot to bootloader command
- A new drive should appear on your computer called **RPI-RP2** or **RP2350** depending on your Pico type.
- Copy the correct **uf2** firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

Firmware configuration

USBSID-Pico has an extensive set of configuration options that expands with each firmware release.

Drivers

Linux Udev rules

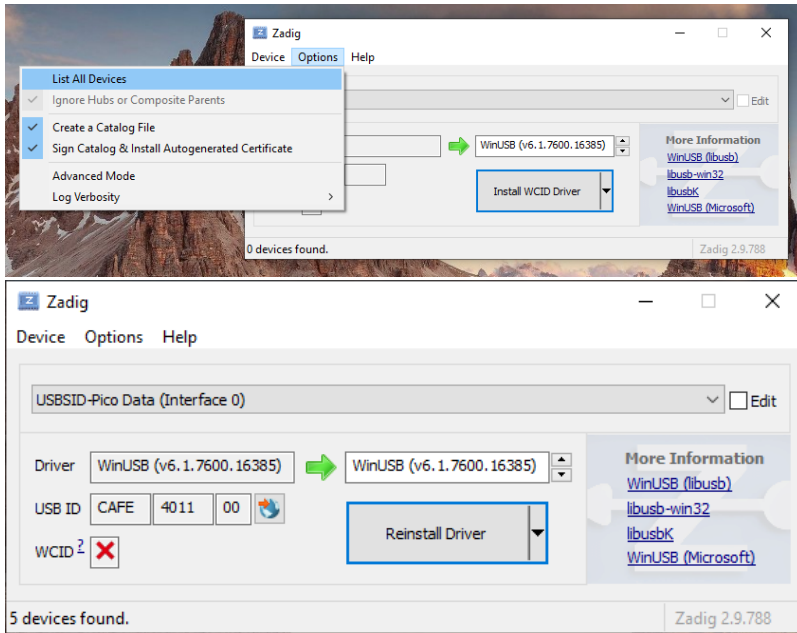
In the [examples/udev](#) directory you can find the [udev rules](#) that I use on Linux.

This purely an example file that you can use and change to your own needs. Steps required for this to work

```
# Check if you are in the plugdev group with the groups command
groups
# If not, add yourself to the plugdev group, then log out and in
sudo usermod -aG plugdev $USER
# Copy the udev ules file to the correct directory
sudo cp 69-usbsid-permissions.rules /etc/udev/rules.d
# Now reload the udev rules
udevadm control --reload-rules && udevadm trigger
# Not working? Try reloading the service
sudo systemctl restart udev
```

Windows driver

Use [Zadig](#) to install the correct driver. Replace the driver for **USBSID-Pico Data** with WinUSB.



USBSID-Configtool

You can configure your board by using the official USBSID-Configtool
The tool is available here [USBSID-Configtool](#) in the releases section

Commandline configuration tool

You can configure your board with: - the commandline config-tool - the commandline config-tool
with the Python Configuration GUI by ISL/Samar if you prefer a GUI.

Python Configuration GUI downloads

[Download script](#)

Python3 → [Download](#)

TkInter → [Installation guide](#)

And either one of the Linux or Windows CLI downloads below

Linux / Windows downloads

The latest version of the CLI config tool is available in a zip file called **tools.zip** from release v0.6.0-BETA and up

Windows downloads

If the DLL's are missing from the zip file you can always download them from here:

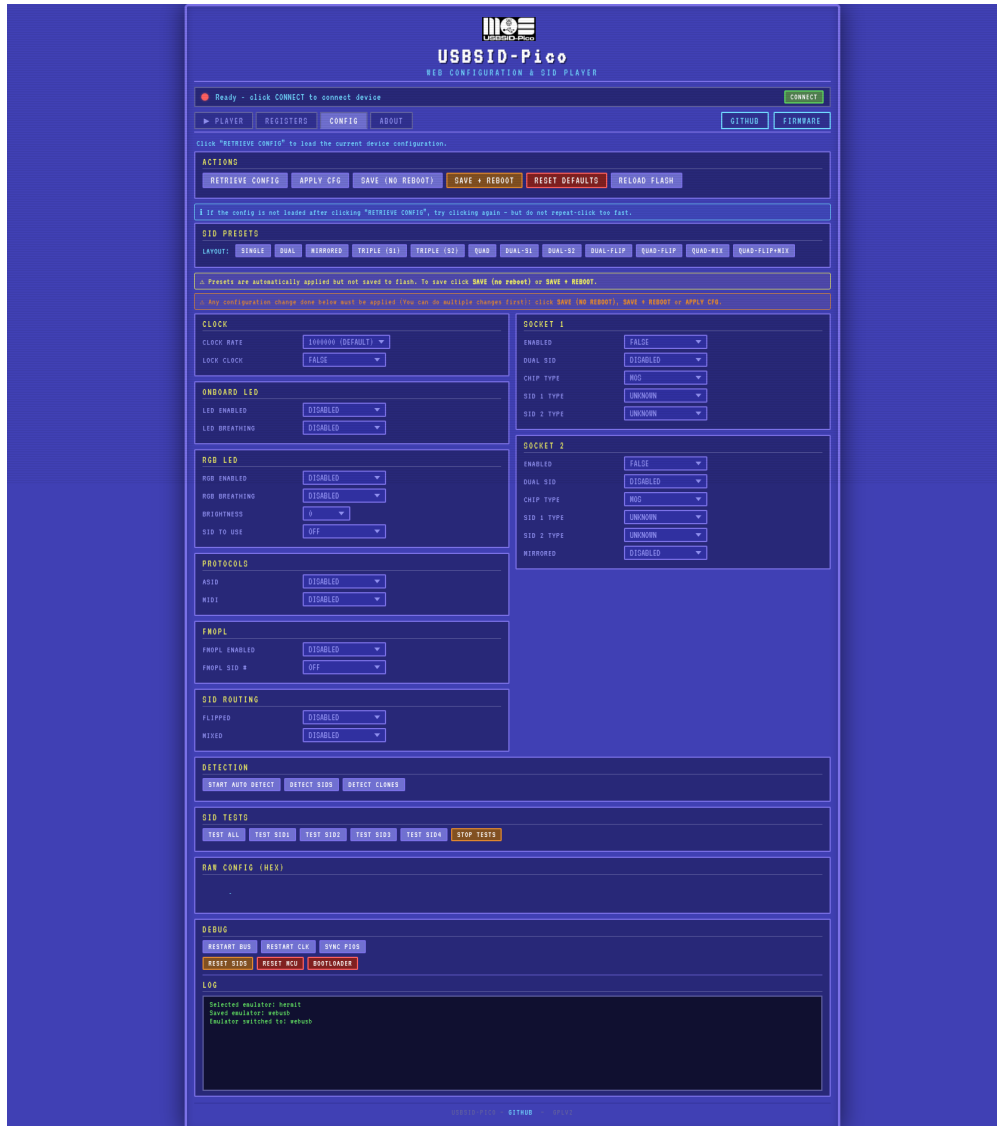
[Download DLL 1](#)

[Download DLL 2](#)

Configuration web page (for Chrome/Edge based browsers using webusb)

USBSID web configuration tool (requires a Chrome based browser).

If needed you can change your USBSID configuration after selecting WebUSB and clicking on [Open config](#).



The player is set up with borrowed code from Deepsid using Hermit's JsSID implementation.

Default configuration (depends on the board version)

The default configuration differs slightly between Pico1 (rp2040) and Pico2 (rp2350) boards.

An explanation for these settings can be found further on in this document.

Board clock rate: **1000000Hz (DEFAULT)** for rp2040, **985248Hz (PAL)** for rp2350

Clock rate locked: **False**

Audio mode: **Mono**

Preset silent auto detection: **False** (**True** for PCB revision v1.5+)

Socket One Enabled: **True**

Socket One Dualsid enabled: **False**

Socket One Chiptype: **Unknown**

Socket One SID 1 type: **Unknown**

Socket One SID 2 type: **N/A**
Socket Two Enabled: **True**
Socket Two Dualsid enabled: **False**
Socket Two Chiptype: **Unknown**
Socket Two SID 1 type: **Unknown**
Socket Two SID 2 type: **N/A**
Socket Two Act as One: **False**
Flip Socket One and Socket Two: **False**
Mix socket addresses (Quad SID): **False**
LED Enabled: **True**
LED Idle breathe enabled: **True**
RGBLED Enabled: **False** (**True** for RGB Pico's)
RGBLED Idle breathe enabled: **False** (**True** for RGB Pico's)
RGBLED Brightness: **0** (**127** for RGB Pico's)
RGBLED SID to use: **-1** (**1** for RGB Pico's)
CDC Enabled: **True**
WebUSB Enabled: **True**
Asid Enabled: **True**
Midi Enabled: **True**
Net Enabled: **False** (**_w** boards only)
NSD Enabled: **False** (**_w** boards only)
FMOpI Enabled: **False**
FMOpI SID no: **0**

Board clock rate

The Commodore64 has different clockrates depending on where you live, USBSID-Pico supports these different clock rates as configuration setting.

Supported SID players will automatically send a new clock rate request to the board to switch to.

Available options are:

DEFAULT 1000000Hz, PAL 985248Hz, NTSC 1022727, DREAN 1023440, NTSC2 1022730

Clock rate locked

Some players start sending data directly after the clockrate is changed. While in most cases this is no issue, some SID clones need some time to start up. This causes issues when the clock rate is changed on the fly.

Also changing the clock rate multiple times after another can cause data corruption to the SID which causes garbled sound.

To overcome this it is possible to lock the clockrate to the current setting it is set to. This blocks the ability for on the fly change requests.

Available options are:

False, True

Audio mode

PCB revision v1.3 only

Sets the audio mode for the headphone jack and audio out pins.

When set to **Mono** the audio of both SID sockets will be combined to mono.

When set to **Stereo** the audio of both SID sockets will be separated into left for **SID1** and right for **SID2**

Available options are:

Mono, **Stereo**

Lock audio switch

PCB revision v1.3 only

Locks the audio switch into its current state (**Mono** or **Stereo**) and prevents it from being changed.

Available options are:

False, **True**

SID Voltage & Configuration confirmation

PCB revision v1.5 only

The board boots with 5v and 9v enabled to automatically detect any changes in the socket / chip configuration, it does this by running a chip and SID detection.

When a change is detected the boards turns off the 9v and 5v and the onboard (RGB)LED starts flashing, making the board unusable until the configuration is validated and confirmed.

Please verify the configuration with care before confirming, failure to do so might damage/kill your SID chips, and we don't want that!

There are multiple ways to validate and confirm the configuration:

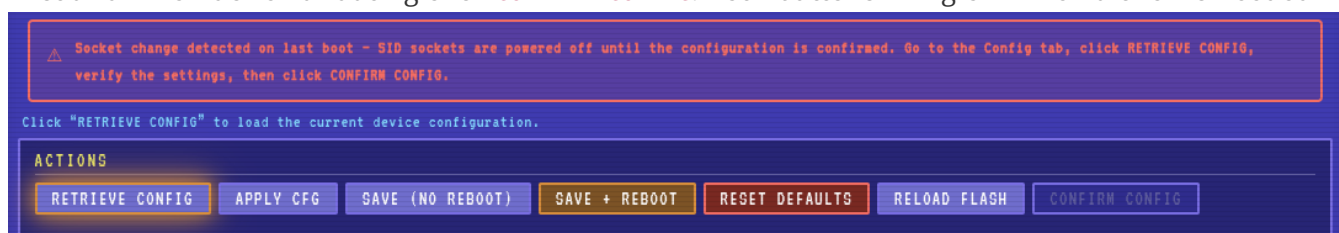
GUI Configtool

On opening the configtool and connecting to the board there will be a text stating the configuration needs to be confirmed. Verify the config on the **SOCKETS** tab and when done click **CONFIRM**



Web Configtool

On opening the config website and connecting to the board there will be a text stating the configuration needs to be confirmed. Verify the config on the **CONFIG** tab by clicking **RETRIEVE CONFIG** first and when done validating click **CONFIRM CONFIG**. Both buttons will glow when a click is needed



CLI Configtool

First read the configuration with **cfg_usbsid -r**, the printed configuration will have a line that states the config needs confirmation. You can do this with **cfg_usbsid -ack**.

Disable socket change detection

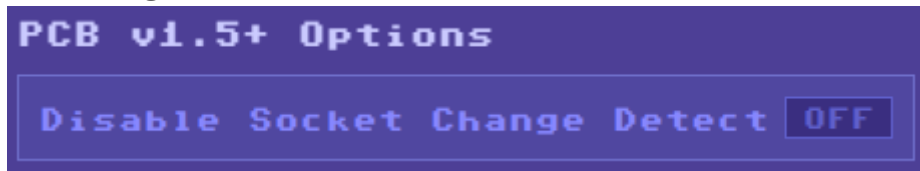
PCB revision v1.5 only

When you have a fixed socket setup, e.g. you do not plan on changing the chips anytime soon, it is possible to disable the change detection on boot. This will make the board start up with 5v and 9v and then - if 12v is required - switch to the required voltage setting during the boot process by switching the voltage back off, setting the correct hv voltage and then switching it back on.

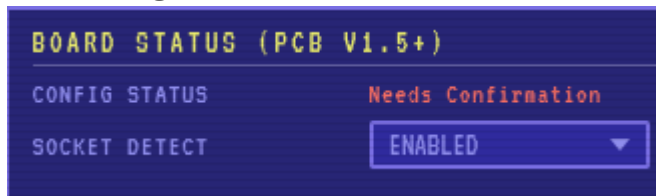
When the socket detection is on, the board will *always* try to detect any changes in the configuration *before* the required voltage is applied.

By setting the **Disable socket detection on boot** option you can control the above behaviour:

GUI Configtool



Web Configtool



CLI Configtool

Change the behaviour with **cfg_usbsid -sad N** where N can be 0 (disabled) or 1 (enabled)

Socket settings

Each socket comes with its own set of configurable options.

Socket One/Two Enabled

Enable or disable the use of this socket.

Available options are:

False, True

Socket One/Two Dualsid enabled

Enable or disable dualsid for this socket.

This requires the **Chiptype** for this socket to be set to any clone chip (not **MOS**) and ofcourse a dual SID supporting SID clone with support of connecting the A5 address line.

Available options are:

False, True

Socket One/Two Chiptype

Set the type of SID chip for this socket.

Available options are:

MOS (real SID), **Unknown**, **SKPico**, **ARMSID**, **ARM2SID**, **FPGASID**, **RedipSID**, **PDSID**, **BackSID**, **SIDEmu**

Socket One/Two SID 1 type

Available options are:

Unknown, N/A, 8580, 6581, FM0p1

Socket One/Two SID 2 type

Available options are:

Unknown, N/A, 8580, 6581, FM0p1

Socket Two Act as One

When enabled, this will disable other Socket Two settings and this socket will receive all writes from Socket One as it were mirrored.

Available options are:

False, True

Flip Socket One and Socket Two

When enabled, Socket One and Socket Two are swapped so Socket One's physical position responds as Socket Two and vice versa.

Available options are:

False, True

Mix socket addresses (Quad SID only)

When enabled, addresses for a 4-SID setup are mixed between sockets instead of being assigned sequentially per socket. Combine with **Flip** for the remaining address orderings.

Available options are:

False, True

Preset silent auto detection

When enabled, the board silently re-runs chip/SID auto detection before applying a preset.

Default is **False** on PCB revision v1.0~v1.3 and **True** on PCB revision v1.5+.

Available options are:

False, True

LED Enabled

Disable or enable the use of the onboard LED.

When enabled this automatically enables the LED as Vu meter for the voices of **SID1**.

Available options are:

False, True

LED Idle breathe enabled

Disable or enable the idle breathing effect for the onboard LED.

Available options are:

False, True

RGBLED Enabled

Only for black Pico clones with onboard RGB LED

Disable or enable the use of the onboard RGBLED.

When enabled this automatically enables the RGBLED as Vu meter.

Available options are:

False, True

RGBLED Idle breathe enabled

Only for black Pico clones with onboard RGB LED

Disable or enable the idle breathing effect for the onboard RGBLED.

Available options are:

False, True

RGBLED Brightness

Only for black Pico clones with onboard RGB LED

Set the maximum brightness of the onboard RGBLED.

Available range is:

0 to 255

RGBLED SID to use

Only for black Pico clones with onboard RGB LED

Set the SID of which to use the voices for the Vu meter.

Available options are (depending on your SID configuration):

-1 off, 0 off, 1 SID1, 2 SID2, 3 SID3, 4 SID4

FMOp1 Enabled

Requires a SID clone with FMOp1 capabilities, the socket **Chip type** set to **Clone** and **SID type** set to **FMOp1** before activation!

Available options are:

False, True

FMOp1 SID number

This setting cannot be changed manually This setting will automatically be set to the correct SID number when all requirements are met as described under **FMOp1 Enabled**

CDC Enabled

This config setting is intended for future use Available options are:

False, True

WebUSB Enabled

This config setting is intended for future use Available options are:

False, True

Asid Enabled

Disable or enable ASID protocol handling (SysEx-based SID playback over MIDI).

Available options are:

False, True

Midi Enabled

Disable or enable MIDI input handling (note/CC/voice control of the SID(s)).

Available options are:

False, True

WiFi Enabled

`pico\_w`/pico2_w boards firmware with _net only

This config setting is intended for future use, no CLI/GUI/Web configtool support yet

Enables the board's WiFi station connection. Requires an SSID and password to have been provisioned first.

Available options are:

False, True

NSD Enabled

`pico\_w`/pico2_w boards firmware with _net only

This config setting is intended for future use, no CLI/GUI/Web configtool support yet

Enables the Network SID Device (NSD) server: a TCP listener (default port 6581) plus a UDP discovery responder, letting NSD clients such as ACID 64, JSidplay2 and SIDPlay find and drive the board over the LAN exactly like a native NSD server. Only one NSD session (WiFi or Bluetooth) can be active at a time; USB always keeps priority over the SID bus.

Available options are:

False, True

Bluetooth NSD Enabled

`pico\_w`/pico2_w boards firmware with _net only

This config setting is intended for future use, no CLI/GUI/Web configtool support yet

Enables the Bluetooth Classic SPP transport for the same Network SID Device protocol as **NSD Enabled** above. No NSD client speaks Bluetooth SPP directly, so this needs a small host-side RFCOMM-to-TCP bridge (see [examples/nsd_spp_bridge.py](#)) to be usable with existing players. Available options are:
False, True

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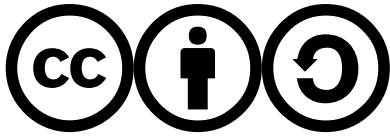
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